

Auteur: Ilyas Sal-lak

Studierichting: Informatica

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Voor $z^4 = -1$ zoeken we de vierdemachtswortels.

$$\text{Formule: } z = \left(\cos \left(\frac{\theta + 2\pi k}{4} \right) + i \sin \left(\frac{\theta + 2\pi k}{4} \right) \right)$$

In poolcoördinaten geldt $r = 1$ en $\theta = \pi$

Voor $k = 0, 1, 2, 3$ vinden we:

$$\begin{aligned} z_1 &= \left(\cos \left(\frac{\pi}{4} \right) + i \sin \left(\frac{\pi}{4} \right) \right) = \frac{\sqrt{2}}{2} + i \frac{\sqrt{2}}{2} \\ z_2 &= \left(\cos \left(\frac{3\pi}{4} \right) + i \sin \left(\frac{3\pi}{4} \right) \right) = -\frac{\sqrt{2}}{2} + i \frac{\sqrt{2}}{2} \\ z_3 &= \left(\cos \left(\frac{5\pi}{4} \right) + i \sin \left(\frac{5\pi}{4} \right) \right) = -\frac{\sqrt{2}}{2} - i \frac{\sqrt{2}}{2} \\ z_4 &= \left(\cos \left(\frac{7\pi}{4} \right) + i \sin \left(\frac{7\pi}{4} \right) \right) = \frac{\sqrt{2}}{2} - i \frac{\sqrt{2}}{2} \end{aligned}$$

De vier vierdemachtswortels zijn dus:

$$\begin{aligned} z_1 &= \frac{\sqrt{2}}{2} + i \frac{\sqrt{2}}{2} \\ z_2 &= -\frac{\sqrt{2}}{2} + i \frac{\sqrt{2}}{2} \\ z_3 &= -\frac{\sqrt{2}}{2} - i \frac{\sqrt{2}}{2} \\ z_4 &= \frac{\sqrt{2}}{2} - i \frac{\sqrt{2}}{2} \end{aligned}$$

References